

Securities Education Certificate



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Lecture 3

Equities

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Accounting

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The main purpose

It's often a grave misconception that accounting is all about finding out how much money a business is *making* in terms of profit. I'd like to take this immediate opportunity to say that is **NOT** what accounting is about. I don't know where this idea has come from over the past 100 years but it's definitely not the main purpose of accounting. It is, however, one of the metrics that accountants use to measure the performance of a company but that is not the preliminary reason in the first place that accounting takes place.

Accounting actually answers a different question..

What is the operational worth/strategy of the business in question?

What do I mean by this? Essentially, the profitability of a business isn't my main interest when I'm looking at a business's accounts. When you are doing accounting, you're essentially trying to track the spending of the company that you're looking at in order to figure out what it's up to at different points in time. What is it spending its money on over the course of a specific month? What revenue streams exist to help money come through the door? How much am I being charged for taxes from His Majesty's Revenue and Customs? These are the sorts of questions that accountants answer because they, er... *account* for the money that's coming in and out of the business over a specific time period.

When I read financial statements, I want to gain an understanding of what the business is up to. Whether or not the business itself is making money is rather irrelevant as long as it's spending money in the right places at the right time. Of course, I'd be alarmed if I was running a venture capital firm and I found out that the founders of the next big social network I'd sponsored had falsely squandered half of the capital on "general expenses" when in fact they'd been seen on social media showcasing their lifestyle in Ibiza surrounded by illegal paraphernalia in the back of a yacht.

This is what accounting does: it shows a total breakdown of the operations of the business and tries to put numbers behind it. Profits and loss figures are only a by-product of all the work done to account for everything that a business does, **NOT** its end goal entirely.

So, now that I got that out of the way, let's discuss some basic concepts in accounting that you'll need to be familiar with.

Cash vs Accrual basis

Cash-based accounting

As I said above, the money-making aspect of a business isn't entirely relevant when you're doing accounting. Instead, you're looking at *what* the business is doing in terms of its operations. This may not make sense to you right now so we're going to try and plug an example instead!

Consider Oscar and I to be running a nice small bakery in the pretty countryside of Devon. We've set up a bakery, QT Bakery and we've started running it from the 1st of January so that our fiscal year counts as January 2022 – January 2023. Because Oscar's a great baker given his 12.5% maternal Swiss background, we've received our first order from a customer today (the 29th of October) to supply £1,000 worth of pastries to their event next week – namely, the birthday of Oscar's mum. However, Oscar's dad (who organised the event) says that he can't pay us until March next year in 2023 because he's waiting for his year-end bonus to come through.

Oscar and I think that we can and will fulfil the order. As we don't have the money right now from Oscar's dad to pay for the necessary £500 ingredients, we've also promised our suppliers that we'll pay those costs the month after Oscar's dad pays us back, which is in April 2023.

So far, so good. This is pretty much standard transactional behaviour in business and nothing shady is happening at all. We are promised money for our work in delivering the pastries and our suppliers are promised money for their supplies once me and Oscar get paid.

But (and this is where you need to realise the problem)... how would this look in accounting if we wrote down the transactions we did? The logical starting point would be to write down what we did in terms of making the pastries and buying our goods alongside how much money you received or gave away for doing those two jobs:

Type of transaction	Value	Transaction Date	Date of payment
Sale of pastries	£1,000	29 th October	March 2023
Purchases of ingredients	£500	29 th October	April 2023

So far, so good. We need to account for this in some way. The common idea would be to look at how much money moved in and out of the business at the time of the transaction date. If we tried this, however, something wouldn't look right...

For the Fiscal Year (FY) January 2022 – January 2023 CASH-BASED	
Account	Value (£)
Sales	0
Costs	0
Profit	0

This isn't a mistake! Remember, we don't get paid until next year. So, in terms of the cash moving in and out of the business, we haven't received any cash from sales or paid for any costs yet, hence our profit of £0. This is otherwise known as **cash-based** accounting as I'm purely accounting for the bakery's operations based on the movement of cash in and out of the business.

But, think about it. If you read that tabulated breakdown above and had no idea of the inner workings of our bakery, you'd believe that nothing was happening in the business, right? You'd think that me and Oscar have

done absolutely nothing at all in that entire year since we haven't made any sales and spent absolutely nada on procuring any ingredients – that's literally that the statement tells you, right? And yet, we have just secured a deal to supply £1,000 worth of pastries on the 29th of October to Oscar's dad and been able to procure £500 worth of ingredients from our suppliers as long as we pay them later!

This is the issue with cash-based accounting: *it does not account for the operations of the business – only the cash moving in and out of the company.* Cash does **NOT** represent what is going on in the business although it is a metric that can be used to measure the business's activity **WHEN APPLIED CORRECTLY!** If you have situations where a business is delaying payment, cash-based accounting is going to mess up the true image of the business's operations in a way that an average accountant or shareholder would think you've done nothing at all in the first place while you've been working unbelievably hard to fulfil orders and the like.

So, we need an alternative system. Enter *accrual*-based accounting!

Accrual-based accounting

Let's instead account for the operations of the business as they happen rather than when we transfer the money for the transactions in question.

For the Fiscal Year (FY) January 2022 – January 2023 ACCRUAL-BASED	
Account	Value (£)
Sales	1000
Costs	500
Profit	500

Ah, this is much more like it. We don't receive the £1000 yet from Oscar's dad nor do we pay the £500 costs of the ingredients but we account for it through an *accrual*-based system, which in essence just means that we take note of it when it happens (i.e. when it is 'recognised') and not when it is earned in terms of the cold hard cash. That way, when someone looks at the business's accounts, they can see the work we've done in supplying the pastries for Oscar's birthday and realise that me and Oscar have actually done something instead of mis-interpreting the lack of cash flowing in or out of the bakery as a sign of inactivity.

We do **NOT** care when the cash comes in if we are doing accrual-based accounting: we care about when the business's operations have taken place! That's the key difference between cash and accrual-based accounting and I insist that this is clearly understood irrespective of whatever you learn in future accounting classes as it forms the basis of how to read and understand financial statements to the best of your abilities!

If you are interested as to when the cash comes in, you'd be more interested in the ideas of *account receivables* in the balance sheet which store information about the incoming money coming into the business alongside the business's *cash flow statement* that tells you exactly what money is flowing in and out of the business at specific times of the year.

Linking the three key financial statements

In essence, you can conclude the work of the three financial statements through its three purposes:

- The balance sheet is a *snapshot* of the business's assets in terms of what it owns and its liabilities in terms of what it owes to any of its debtholders and its owners.
- The income statement links two balance sheets to show how the change in the value of its total assets was achieved between two different time periods.
- The cash flow statement links two balance sheets to show how the change in the total value of the business's cash reserves was achieved.

By keeping those three points in mind, you'll be able to read any financial statement and ensure that you can invest appropriately, write effective yet insightful equity research reports and deliver DCF models with intuition and clear clarity since you'll know exactly what you're doing in terms of the three key financial statements and why they work the way they do.

Technicals

Some of you who are reading this definitely attended Justin's accounting workshop the other week over the course of the two sessions. While I have to admit that it was a great workshop, there were those amongst you who seemed a bit confused about some of the terminology (not to mention the technical issues that Huxley and the college's I.T. department seem to continuously exhibit whenever a large event is being run in South Kensington).

I thought it would be useful to cover a few of these terms for the interests of those of you who are a bit stuck on what they mean and how they may make more sense by thinking of them from an alternative perspective. First, a discussion on EBIT and its sister, EBITDA.

EBIT and EBITDA

Consider a simple income statement like the one you see below:

Income Statement (£000s)	
Revenue	1000
Cost of Goods Sold (COGS)	(800)
GROSS PROFIT	200
SG&A	(90)
Depreciation	(10)
OPERATING PROFIT	100
Interest	(10)
INCOME BEFORE TAXES	90
Tax (30%)	(27)
NET PROFIT	63

Here, the net profit of the business from the income statement above was £63,000. This is how much the company *earns* after every single possible cost has been deducted from the revenue.

EBITDA is simply the earnings before interest, taxes, depreciation and amortisation. The formula you normally get given is something along the lines of:

EBITDA = Net income (profit) + interest + taxes + depreciation + amortisation

Here, all we're doing is we're going from the bottom of the income statement (i.e. the net profit) and we're working our way back up the income statement to get the earnings of the company before the ITDA deductions were accounted for. It's pretty simple:

$EBITDA = 63 + 27 + 10 + 10 (+ 0) = 110$ (i.e. £110,000). That's basically the value of the earnings after we've deducted SG&A from Gross Profit. That's it, honestly!

EBIT is even simpler. Forget the depreciation and amortisation section:

EBIT = Net Income (profit) + interest + taxes = $63 + 10 + 27 = 100$ (i.e. £100,000). That's interesting...

That's literally what the operating profit signifies. So really, EBIT is basically operating profit if we work our way from the bottom of an income statement and travel upwards, reversing several deductions that we did from the top to end at the value of net income.

Now, why do people care about EBITDA and EBIT so much? Well, think about it. Operating profit tells you a rough sense of how much money the business is generating *before* you worry about the more irrelevant stuff including taxes and interest. The latter two aren't *really* going to stain the quality of the business's operations are they? So, operating profit gives us an idea of how much raw hard cash the business is making.

However, when Justin alluded to the mechanics of depreciation and amortisation, all we're doing is spreading out the cost of an asset's purchase over the time that we use the said asset. We've already spent all our hard earned cash buying that asset but we're doing an accounting trick to show that it's being used over a fixed period of its useful life. That does **NOT** mean that we're losing money through depreciation at all! All we're doing is saying "We've bought this car for £20,000 cash but we're going to say in our accounting books that the car cost £5,000 each year for 4 years since that's how long we use it for and that's how much value it loses every year." The car's already used up £20,000 of your cash stores back when you first bought it – you're not losing any more money on its purchase over the time that you actually use it!

For that reason, we add depreciation and amortisation costs as the business isn't actually losing any money after the purchase on the asset itself – it's simply spreading out the cost appropriately to account for its cost as a function of how long it's going to be useful before it gets thrown away. To really add to the picture, you may see depreciation being included as a Cost of Goods Sold (CoGS) so be careful as to where you see depreciation in the income statement that you've got in front of you.

As for EBIT, it only helps us understand how the business is performing without considering costs unrelated to the goods or services that it sells. Remember, interest is dependent on the amount of money raised by the business through debt and taxes are dependent on where the business is located. Neither of the two have anything to do with the business's main good/service-selling operations so EBIT and EBITDA show the business's earning potential only through its main operations in a clearer sense.

Enterprise Value

This term isn't as hard as it looks. Essentially, when you're looking at a business, you're aware that the business itself has assets that help it perform its operations. A business can raise money in two ways:

1. Issue debt. It can e.g. take out a loan to go and buy its equipment and all the material it will need to produce its goods and services.
2. Issue shares to sell equity. It can sell shares of the company to interested people so that they can become shareholders. **NOTE** even if you launch your own company with no other shareholders, you've still issued shares to yourself in the company (i.e. all 100% of the company's shares) in return for your own money, so the definition holds true even for a sole proprietor.

So, the combination of debt + total value of shares (i.e. market cap) is how the company sources its funds to run its operations.

However, here's a hypothetical for you to think about. If a company takes out a loan of £1 million and sells shares to the tune of £1 million, it raises £2 million in total. If it stores that entire £2 million as cash which sits in the company's vaults, that £2 million in cash isn't actually contributing to the company's operations. If the company's main planned operations is building high-end computers and all it has so far is just a cash reserve of £2 million, that £2 million in cash is literally completely useless to its main aim of building high-end computers. As a result, the company's *value of operations* is literally zero. Hell, if the CEO actually spent all £2 million of it to build factories and source material, then its value of operations is exactly £2 million as that's literally how much it cost to get the business's main task of building computers actually going.

What enterprise value really measures is the value of the company's operations. To do this, it combines the value of the shares it's sold alongside the debt it's raised and removes any cash that doesn't play an active part in its operations. Hence, the formula for Enterprise Value (EV) is:

Enterprise Value = Market Capitalisation + Total Debt - Cash.

Alternatively, you might see some people mention EV as a way of seeing the amount of money it takes to buy out the business entirely. This is also intuitive through a similar analysis: you have shareholders who want to be paid the market rate for their shares (market capitalisation) and people who are owed money due to the company's debt (total value of debt). Once you paid them both off, you then have the company to yourself and you're rewarded with some of the cash reserves to offset the size of your purchase. The two explanations (although different) provide the same final formula in terms of what enterprise value actually stands for.

Market Capitalisation

This is, er... a badly explained term normally in financial academia. It's actually quite simple to understand so I'm not sure why people get this term so badly confused. Market capitalisation is the value that the *market* gives to the value of shareholder's equity. This is achieved by taking the market price of each share and multiplying it by the number of shares that the company has issued.

Market Capitalisation = Share price × No. of issued shares

Why is this relevant, you might ask? Companies get their accountants to value the assets that they own, which in turn calculates the value of equity that shareholders are entitled to. However, accountants may value the business's assets at a value greater than or less than what the market values it at. Market capitalisation is the valuation of the shareholders' equity (and in turn, the assets of the business) from the perspective of market

players. If the market values the shareholders' equity at a value *greater* than the business does, there might be an intangible asset e.g. brand value that the accountants haven't been able to account for. If the market value of shareholders' equity is *less* than the business's valuations, the accountants have clearly overvalued their business and something weird is going on inside the business's accounting and valuation process.

Simple, really. Nothing too complicated there.

The WACC

Let me first spell out what it means: the Weighted Average Cost of Capital. The problem with teaching this formula is that practically no-one (and I mean literally no-one) that I've met has a single clue what it means. Often, you'll see this formula:

$$(E \times e) + (D \times d \times (1-T))$$

Where:

E → cost of equity
 e → market value of equity
 D → cost of Debt
 d → market value of debt
 T → tax rate

That's all fun and games but it means absolutely nothing at all if you don't know where it comes from. Evidently, this formula wasn't developed by some imaginary maths fairies in Canary Wharf or on Wall Street.

So, since I'm a chemistry student and chemists care about where formulae and theory come from rather than just blindly apply it (as you might blow something up), let's try and derive it.

First, let's discuss the idea of a weighted average.

What does the Weighted Average actually mean?

Let's calculate the mean of 5 numbers: 1, 2, 3, 4 and 5. I'm sure that given most of you did GCSE maths, this won't be too hard:

$$\frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

Thanks to the properties of fractions, we can write this in an alternative form:

$$\frac{1}{5}(1) + \frac{1}{5}(2) + \frac{1}{5}(3) + \frac{1}{5}(4) + \frac{1}{5}(5) = 3$$

Here, we've basically multiplied each value by $\frac{1}{5}$ and totalled the result. One way you can consider this is that we've *weighted* each number by $\frac{1}{5}$ and calculated their total sum. **This is a weighted average.** Each number in the list has a weight of $\frac{1}{5}$ that we use to multiply each number and add together to calculate the value of the mean.

However, we might wish to change the weights if we have a scenario where one number has more importance than another e.g. with exam marks.

I take three exams, both out of 60 marks. I scored 30, 40 and 50 in all three exams. If all the exams had equal importance (i.e. 33⅓% each), I averaged 40/60.

However, the third exam (the one that I scored 50/60) actually counts for 50% of the weight of my overall mark, with the other two exams only counting 25% each. This means that the third exam is more important than the first two exams. How do I calculate my weighted average mark?

This isn't actually that hard. We know the total percentage is 100%, of which 50% is allocated towards the second exam. Adapting the example above, we get:

$$\frac{25(30) + 25(40) + 50(50)}{100} = 42.5$$

So, I actually ended up scoring 42.5/60 thanks to the fact that the exam which counted more (i.e. the third one) actually contributed more to my final mark, an extra 2.5 marks more. Fantastic!

Similarly, if I know what the total sum of the weights are (in this case, 100), I can calculate decimal values instead of writing a fraction:

$$0.25(30) + 0.25(40) + 0.5(50) = 42.5$$

There is also a mathematical description using summation to help me simplify the notation of the weighted average but since I'm not a mathematician, I'm not really bothered.

The relevance towards the WACC

I've already mentioned the idea of the weighted average (i.e. the 'WA' of the WACC).

What do I mean by the cost of capital? Do you remember how companies raise funds to kickstart their operations? They issue shares to potential shareholders, which is the company's **cost of equity** and they also take on debt from third-parties, which is their **cost of debt**. Both these two contribute to the total **cost of capital**, which is how much it costs the business to run its operations. There's another definition as well but we'll come onto that momentarily.

Remember, the cost of capital isn't just the normal mean of the cost of equity and cost of debt because they may not be allocated in equal amounts. Think of it, if 25% of the company's capital comes from shareholders buying shares and the other 75% comes from raising debt, it's a pretty silly idea to then work out the *average* cost of capital by simply adding the two numbers up and dividing by 2 (we divide by 2 because there's only 2 ways of raising capital) since we'd be weighting both costs of capital by 50% each ($100 / 2 = 50$), essentially saying that the cost of equity counts more and the cost of debt counts less towards our final average.

So, if we want to work out the average cost of capital, we'd take a *weighted* average. For the example above, we'd do:

$$\frac{(25 \times C_E) + (75 \times C_D)}{100} = WACC$$

More generally, we could work out our final weight by adding both the cost of equity C_E and cost of debt C_D . Then, We could weight each term accordingly based on how much they contribute to the total sum of C_E and C_D . Writing this out, we get:

$$\left(\frac{E}{E + D} \times C_E \right) + \left(\frac{D}{E + D} \times C_D \right) = WACC$$

We're getting closer! Sometimes, the combined value of equity E and debt D gets fused together to give another term, V . So, sometimes you see:

$$\left(\frac{E}{V} \times C_E \right) + \left(\frac{D}{V} \times C_D \right) = WACC$$

Some of you might have wondered '*isn't E and C_E the same? They're both the cost of equity, aren't they? Aren't D and C_D the same thing here?*' And if you thought that, you'd be right! Sadly, because some academic people in finance do things their own way, they've sort of forgotten the rules of maths and because of these rules (which I'm discussing in the next paragraph), I've had to separate the two terms accordingly because I'm doing to do something of a Legendre-transform-esque magic trick.

There's one thing I haven't explained to you yet... What is the cost of equity? What does the cost of selling shares mean? What is the cost of taking on debt? For the latter, you'd know what that is... it's the **interest rate**! It's a percentage of the loan that you have to pay back on top of the original loan because of the fact that you've rented money from a third-party. Read this bit very carefully:

The cost of selling shares is that shareholders expect a percentage return on the money they invested in buying your shares! The cost of taking on debt is that the debtholders expect a return on the amount of money they've given you, which is the interest rate.

The shareholders might want to buy a share of your company and expect you to pay back around, say... 5% a year of their investment. Likewise, the people who you owe debt to might expect you to grow the company in a way that they expect you to pay back around, say... 7% a year on their loan that they gave you. What's the average rate of return from both the shareholders and the debtholders? Is it slap bang in the middle at 6%? No. Why? Because you have a disproportionate amount of cash raised from equity shareholders and debtholders in your company. In our example, 25% of the capital raised came from shareholders whereas 75% came from debtholders so it'll be closer to 7% than 5% (it's 6.5% if you do a weighted average).

For this reason, the cost of equity and the cost of debt are the percentage returns that the shareholders and the debtholders expect from buying your shares or lending you a loan. So, the WACC actually gives you a proportionate value of the expected return from your business. I prefer to now complete the job by changing C_E to R_e (return on equity) and changing C_D to R_d (return on debt) to match what's often taught on the CFA Level I exam:

$$\left(\frac{E}{V} \times R_e \right) + \left(\frac{D}{V} \times R_d \right) = WACC$$

Sometimes, you'll see a version of the formula that includes the tax rate, t :

$$\left(\frac{E}{V} \times R_e\right) + \left(\frac{D}{V} \times R_d\right) (1 - t) = WACC$$

It seems fairly strange that the tax has a part to play in the calculation for the WACC. What do taxes have to do with this (if anything at all)?

Recall the income statement earlier:

Income Statement (£000s)	
Revenue	1000
Cost of Goods Sold (COGS)	(800)
GROSS PROFIT	200
SG&A	(90)
Depreciation	(10)
OPERATING PROFIT	100
Interest	(10)
INCOME BEFORE TAXES	90
Tax (30%)	(27)
NET PROFIT	63

Tax is paid *after* the interest is paid for. So, if you have a lot of debt, the amount of interest you pay increases **BUT** the amount you pay through tax decreases! *In essence*, debt reduces the amount of tax you owe by lowering the income before taxes. We say that interest is a tax-deductible expense as it is deducted from the total amount of tax that you have to eventually pay! Sometimes, you might read this term being named 'tax shield', which is what the WACC accounts for. In essence, debt actually does have a benefit!

Here's a classic example to get things in perspective. Consider that I earn a salary of £100,000 (very hypothetical and unlikely, but it'll do). If the flat tax rate for my salary is around 30%, I end up having to pay £30,000 in taxes to Her Majesty's government and I get to take home £70,000.

What if I had *also* taken out a loan of £200,000 and have to pay 10% interest on it every year? Well, I have to pay £20,000 in interest. **BUT**, what do I pay in taxes? I don't get taxed on the £100,000 that I take home – remember the income statement above. Interest is tax-deductible as it reduces the net profit I get to take home.

If I have to pay £20,000 interest from my £100,000 salary, I am only taxed on the remaining £100,000 - £20,000 = £80,000. As a result, I pay 30% on £80,000, which is £24,000.

Wow.

By taking on some debt, I've actually reduced the amount of tax that I pay from £30,000 to £24,000. Yes, I would only be taking home £56,000 but that's an entirely different story altogether. Here, I'm bothered about the benefit of the debt and not how much profit I'm making. In essence, the debt of £200,000 has actually saved me £6,000 a year! As one great quote I read on this subject read: '**money saved is money earned**'. By saving £6,000, the debt has helped me actually earn £6,000! So, really, the debt I've taken out hasn't cost me £20,000 overall but instead actually costs me £20,000 - £6,000 (cost - gain) = £14,000.

So, the interest cost is actually £20,000 - 30% (i.e. 30% on the £20,000) = £20,000 × [1 - 0.3] = £14,000.

More holistically, if a debt costs me 10% of my returns but the tax rate is 25%, the actual cost of my tax is:

$$10\% \times [1 - 0.25] = 7.5\%$$

Phew. That's a very hard concept to understand and often one that is rarely mentioned. I hope that does add a clear image on the WACC and where it comes from so that in future, you know exactly what you're trying to calculate when you're doing DCF calculations, for example.

Capital Asset Pricing Model

There might be situations where you won't have the value of equity to hand or you might have to calculate it by hand. How do financial academics do this? They have a model (just like they do for literally anything) called the Capital Asset Pricing Model.

It's a bit hard to explain this bit without explaining what the government bond is. So, sometimes you'll hear some terms that you'll familiarise yourself with in Lecture 6. For now, I'm just going to pre-emptively mention a few of them with a little bit of context so that you can understand the next few points. Once you've completed Lecture 6, you can return to these notes and fully understand the ideas that I've discussed.

The trouble with buying stocks is that it's risky. Ah, great – another term you'll find more about next lecture. If something in finance is risky, it means that it can make you lose money. If I buy a share of a company, I might lose money as the value of that share can go down to £0 just as much as it could go up in value. The safest thing that I could buy into would be a government bond because which government could never pay back its bondholders? If it runs out of money, it could easily increase taxes to recoup the money owed.

In economics, if I invest in something with my money, I've lost out on other opportunities to invest into other things. I can measure my loss in opportunities by measuring the **opportunity cost**. When buying a stock, this opportunity cost is the value of a government bond that I could have invested in. The riskier the stock that I buy, the greater the potential returns. This comes down to whether the stock price follows the pattern of the entire stock market. If the stock is very risky, it behaves erratically and unpredictably **BUT** this could also mean I could make a lot of money while everything else behaves predictably – this is the essence of stock trading: finding stocks that could be very risky compared to its competitors but also provide huge potential returns.

In theory, I could define my risk in terms of my opportunity cost through a value denoted as β , where the higher the β , the higher the risk. I could define a zero-risk investment (i.e. $\beta = 0$) as an investment in a government bond and define a straight line relationship between the riskiness of my investment and the potential returns I get. **This is the Capital Asset Pricing Model.**

Putting it in equation form (just like $y = mx + c$), it gives us:

$$R_a = R_f + \beta(R_m - R_f)$$

where:

R_a = Expected return on an asset (i.e. the share) in question **GIVEN BY THE MODEL**

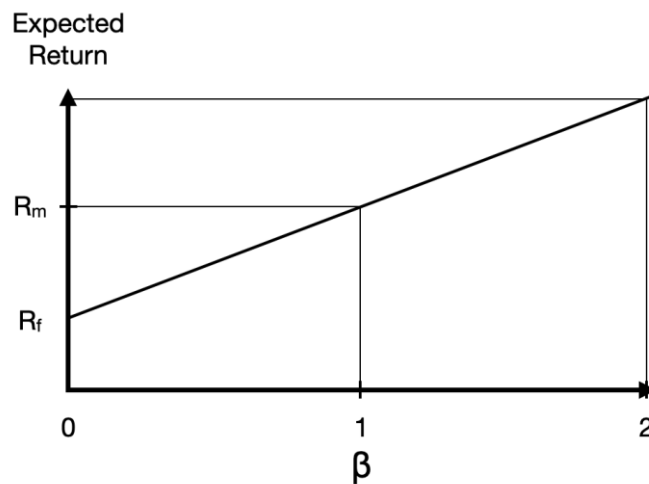
R_f = Return on a risk-free investment

β = The 'riskiness' of the asset (i.e. the share) in question

R_m = Return given by the market

Normally, some economists like to mention that the $R_m - R_f$ term is the 'market-risk premium', which essentially just stands for how much you're willing to pay more in the market for the risk you're taking on as a share buyer.

Here's a nice little illustration of the CAPM to show you how this looks:



If I'm honest, I've never really liked the CAPM. Primarily, my issue is that the maths tries to *predict* the future through using past data. However, past information is never indicative of future success. Just because Barcelona won 4 Champions League titles in the last 18 years does **NOT** mean that they'll win one in the next 10 or 20 years. Maths does use probability to do its analysis, although probability is something that will never be able to predict events with absolute certainty i.e. a probability of 1. Of course, mathematicians have very little choice in the data they can use otherwise so they're forced to use historical data. But, the strange yet blind belief that quantitative modellers know precisely how the market behaves and have developed safeguards in their models is a complete fallacy that shouldn't be believed at any cost!

There are other professional issues that the CAPM struggles with. Firstly, the idea of a risk-free rate. We define the risk-free return rate as the rate given by government bonds with zero risk. This does **NOT** hold true! If a government can (and sometimes has) collapsed, the 'risk-free' returns don't become 'risk-free' any longer as you could lose money by realising that the government can't pay you back.

I haven't mentioned this but β is actually based on the historical risk of a stock in relation to the market. For a β of 1, the stock behaves just like any other stock you could find on the market hence why we denote the market return R_m for a β of 1. For a β of 0, the stock does not behave like anything on the market and is essentially mimicking the risk-less nature of a government bond.

Historically, *some* markets have behaved normally **BUT** this does not mean that they'll continue to behave the same in the future. Just like a gambler's fallacy is that they think they can carry on winning whilst they're on a streak before they lose the next game and it's all over for them, people in markets are reminded that 'black-

swan' style events are very possible and if they're not factored appropriately in mathematical models, it's very easy to ruin yourself by not hedging the risk of losing money in an appropriate fashion.

You'll come across this CAPM idea again if you choose to study my other lecture course next term, the Imperial Blockchain Quantitative Trading course. **Be sure to register your interest during the lecture or using the form link below!**



Sign up to Imperial Blockchain's Quant Trading Series, coming in January 2023.
You don't want to miss it (no, really... YOU DON'T want to miss it!)

Right, that's all I have to say on some of the accounting stuff that Justin's discussed. If you're interested in having another accounting session delivered by me this time in the style I've presented the SEC content so far (i.e. with intuition and relevant derivations), you're best off speaking to myself or Ryan about this at the end of a lecture.

Otherwise, happy accounting!

Equities

Notes by **Adam Snelling**

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Asset Classes

An asset is any item of economic value, this can range from anything as simple as a car or a watch, to something as complex as an options contract.

Asset classes are ways in which investors group different assets that have a similar financial structure and are subject to similar laws and regulations.

Focusing on different asset classes can make it easier for investors to diversify their portfolios and spread their risk.

What Are the Most Popular Asset Classes?

There are many options when it comes to investing in different asset classes. Historically, the most popular asset classes have been equities (shares in companies), fixed income (bonds) and cash equivalents. However, as the financial world has become more complex and automated, we have more complex investment options.

These range from financial derivatives (futures, forwards etc) to more recently, cryptocurrencies. Hence there is a plethora of choice for the opportunistic investor. Typically, each asset class also has a level of risk associated with it, for example:

Fixed Income (e.g. Bonds)	Equities	Commodities	Cryptocurrencies
Low Risk	Medium Risk	High Risk	High Risk

We will be explaining why Equities are seen as a relatively risky investment, but also different ways investors trade in the stock market to reduce that risk.

What are equities?

Equity represents ownership in a company. Let's assume you go to the stock market and buy shares in a company called ABC. Once you have executed that trade you are now a part-owner of ABC.

What does this mean? Well, if the value of ABC's shares go up you can sell your particular share back on the stock market for a profit. Likewise, if the shares go down, you will sell at a loss. You might also be entitled to dividends (we will go through what this means later) and depending on what percentage of all shares you own you could have influence on the company's operations.

What affects equities?

If I could tell you exactly what affects the value of a company's shares, I probably wouldn't be writing these notes. Anyway, what you need to know is just like any investment vehicle, equities are affected by a large number of complex factors that make it difficult for any individual or institution to predict.

Like any asset, the price of a company's shares depends on supply and demand for that stock at the time. However, there are ways in which people try to assign a value to a company and its stock and we will have a look at this later.

Private Equity

Private equity refers to company stock that is not publicly listed on a stock exchange. Investing in private companies is quite different from public companies as there is usually a high barrier of entry due to the large sums of money required to invest in these companies. This is a popular investment class for high net-worth individuals due to the possibility of high returns. However, there are ways for individual investors to get involved with private equity.

Private equity firms specialise in investing in private companies before they go public. They can earn money by charging management and performance fees from investors in a fund. Private equity capital can be utilised to fund new technology in emerging markets, make acquisitions of other companies, and expand working capital for a business.

Public Equity

Public equity is much easier for individual investors to invest in as you can just purchase the company's shares on a stock exchange. There is no limit on how much you can invest, and there is greater liquidity than private equity, as you can sell your shares back onto the stock market whenever you like.

Both public and private equity have advantages and disadvantages for companies and investors.

One of the biggest differences between private and public equity is that private equity investors are generally paid through distributions rather than stock accumulation. Distributions are just ways of returning investment gains amongst investors, whilst stock accumulation refers to increasing the amount of stock someone has in a particular company.

Stock Exchange

Now, with all this talk about public and private equity, you might be wondering how you actually purchase equity for yourself. Well, let me introduce to you the stock exchange.

An exchange is just a place where buyers and sellers meet to *exchange* goods. In the past this used to be a physical place where buyers would place their bids for a good, and sellers would place their offers for said good, hence creating a market where buyers and sellers can agree on prices.

This philosophy is maintained in modern exchanges, only with this market-making process being a lot more automated and efficient due to the use of computers. Publicly-listed companies are traded in the UK mainly via the London Stock Exchange (LSE) and it is likely where you would go to purchase company stock.

Let's assume that you would like to purchase shares in ABC, in the past you might have been able to just walk into the LSE and start bidding your price, however with larger swathes of investors and heightened security, this is not possible. Hence, you must place your order through what is known as a broker.

A broker has financial and legal right to trade in the London Stock Exchange, and hence will execute a trade on your behalf. The broker buys the shares from the exchange using the money you have supplied them and then returns the shares to the account you have with the broker.

Understanding the London Stock Exchange (LSE)

You might have heard of something called the "Big Bang". This refers to the government's deregulation of the London stock market in 1986, an event that led to a modernised electronic trading system and opened up the LSE to capital markets worldwide.

The immense deregulation of the financial markets was introduced in a bid to establish London as a hub of financial services and encourage free-market competition.

This enabled the London Stock Exchange to become one of the most international stock exchanges, rivalling other stock exchanges around the world such as the New York Stock Exchange (NYSE).

IPO

An initial public offering (IPO) is the process by which a private company becomes publicly-listed on a stock exchange, allowing anyone to trade its shares. The companies must meet requirements specified by that specific exchange.

IPOs allow companies to gain further capital investment by offering their shares to a wider audience of potential buyers. It can also be seen as an exit strategy for the company's founders as well as those who invested when the company was still private. Realising the profit from their investment.

How an Initial Public Offering (IPO) Works

Before an IPO, a company is considered private. As a pre-IPO private company, the business has grown with a relatively small number of shareholders including early investors such as the founders along with professional investors such as venture capitalists.

Before the IPO the company will release what is called a Prospectus. In this document the company reveals all the details relating to the release of their shares onto the secondary market (stock market). This is usually done with the aid of an investment bank, where the number of shares and price of each share are included, as well as any other business information.

Investors can then apply to buy the shares while the offer period is open. Once investors have been allocated their shares, the stock is now free to trade on the stock market the company is listed on.

One of the main reasons for issuing an IPO is the raising of capital. As a result of offering shares in the company to the public, capital can be accumulated at a faster rate which can lead to greater growth in the company if invested properly.

There are also other legal obligations that come with being a public company as compared to a private one. This is mainly to do with consumer protection regulations. As a result of this greater transparency, companies are likely to experience more favourable borrowing terms due to less risk on the behalf of the creditor, leading to a lower cost of debt.

However, there are still some unpleasant circumstances that come with becoming a public company. The most prominent of these are the increased costs that come with an IPO, most of them unrelated to the cost of doing business. This can include legal and accounting fees on required processes before the company can be listed. However, by far the greatest of these costs are the underwriting fees, where an investment bank will underwrite the value of the shares about to be listed on the stock market. Charging up to 7% of the value of the shares being listed.

Another potential disadvantage could be the distraction of the price of the company's stock, which could be influenced more by factors outside of the company's control. However, this could lead to fragmentation in the company's goals and objectives if the wrong things are prioritised.

The context of accounting

Balance Sheet

A balance sheet is a financial statement that reports a company's assets, liabilities, and shareholder equity. You might be asking what a liability is, well it is crucial to understanding how businesses. Liabilities should not be confused with expenses, expenses are the costs a company generates from its operations, whilst liabilities are the debts the company owes.

Say ABC just received a loan from a bank of £1000 to help them get up and running. An example of their expenses would be their employees wages, but a liability would be the debt plus any interest they owe to the bank.

Fundamental analysts use balance sheets to calculate financial ratios that can help give an idea of what a company is worth.

Balance Sheet

ABC Limited

Q1 2022

All values in USD \$

Assets

Current Assets

Cash	1,000,000	
Accounts receivable	200,000	
Inventory	300,000	
Short-term investments	-	
<i>Total current assets</i>		1,500,000.00

Non- current Assets

Long-term investments	2,000,000	
Property, plant, and equipment	500,000	
Intangible assets	500,000	
<i>Total non-current assets</i>		3,000,000.00

Other Assets

Deferred income tax	200,000	
Other	-	
<i>Total Other Assets</i>		200,000.00

Total Assets	4,700,000.00
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Liabilities and Owner's Equity

Current Liabilities

Accounts payable	300,000	
Short-term loans	400,000	
Income taxes payable	200,000	
Salaries and wages	300,000	
Unearned revenue	300,000	
<i>Total current liabilities</i>		1,500,000.00

Non-current Liabilities

Long-term debt	1,500,000	
Other	200,000	
<i>Total non-current liabilities</i>		1,700,000.00

Owner's Equity

Owner's investment	1,000,000	
Retained earnings	500,000	
Other	-	
<i>Total owner's equity</i>		1,500,000.00

Total Liabilities and Owner's Equity	4,700,000.00
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How Balance Sheets Work

The purpose of a balance sheet is to provide a snapshot of a company's financial position at the time. However it can be used in different ways by different parties.

When a key stakeholder in the company or an employee reviews the company's balance sheet it gives an idea on whether the company is succeeding or failing to meet its targets. However, when the balance sheet is reviewed externally, for example by potential investors, it gives insight as to the health of the company and allows them to assess whether the company could be a good investment opportunity.

The balance sheet adheres to the following accounting equation, with assets on one side, and liabilities plus shareholder equity on the other:

$$\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$$

In other words, the balance sheet must always be *balanced*. This formula is intuitive. That's because a company has to pay for all the things it owns (assets) by either borrowing money (taking on liabilities) or taking it from investors (issuing shareholder equity).

Assets

We already know what assets are, but we can see how they are shown on a balance sheet.

They are listed from top to bottom in order of their liquidity. This is the ease with which they can be converted into cash. They are divided into current assets, which can be converted to cash in one year or less; and non-current or long-term assets, which cannot.

Current assets can include:

- Short-term investments
- Advance payments

Non-current assets include the following:

- Long-term investments
- Intangible fixed assets (such as a patent)
- Tangible fixed assets (such as property or equipment)

Liabilities

Now that we know what a liability is, we can see that on a balance sheet liabilities are divided into similar sections as assets, with current liabilities being those due within one year, and long-term or non-current liabilities are debts due after one year.

Current liabilities might include:

- Taxes due within a year
- Short-term loans
- Money owed to suppliers

Long-term liabilities could include:

- Long-term loans
- Bonds payable

Bonds payable refers to the amount of money the company will owe people that have bought their issued bonds.

Shareholder Equity

Shareholder equity refers to how much of the company is owned by shareholders once liabilities have been subtracted from assets. It is essentially the money belonging to the business shareholders as a result of investing in the company, note this includes business owners as well.

Limitations of a Balance Sheet

There are some issues that come with using a balance sheet, similar to all other financial statements. One thing to watch out for is the valuation assigned to certain assets. These can sometimes be quite inaccurate as the valuation could be quite old and hence not represent current market value. Therefore, it might be worth doing your own personal research into said market, rather than taking the value on the balance sheet as said.

Not only might values be unintentionally inaccurate, but some companies might deliberately alter the values on the balance sheet in order to make the company look more favourable to potential investors. This is particularly true for intangible assets where they are more difficult to quantify their value. For example, a company might include some fictitious assets such as discounts in costs of goods to inflate the value of the company.

Cash Flow Statement

A cash flow statement is a document that records aspects of a company's operations. It is usually released at the end of every quarter for the previous quarter (three months) and does exactly what it says on the tin: describes the flow of cash.

What does this mean in practice? Well we can break down the cash flow statement into main components:

1. Cash flow from operating activities
2. Cash flow from investing activities
3. Cash flow from financing activities

Cash Flow Statement

ABC Limited

All values in USD \$

Q1 2022	January	February	March
Operating Cash Flow			
Net Earnings	200,000.00	150,000.00	175,000.00
Depreciation & Amortisation	50,000.00	50,000.00	60,000.00
Taxes	10,000.00	5,000.00	6,500.00
Net Cash from Operations	140,000.00	95,000.00	108,500.00
Investing Cash Flow			
Investments in Property	50,000.00	30,000.00	30,000.00
Net Cash from Investing	50,000.00	30,000.00	30,000.00
Financing Cash Flow			
Repayment of debt	100,000.00	150,000.00	125,000.00
Repayment of equity	50,000.00	45,000.00	55,000.00
Net Cash from Financing	-150,000.00	-195,000.00	-180,000.00
Net change in Cash	40,000.00	-70,000.00	-41,500.00
Opening Cash Balance	50,000.00	90,000.00	20,000.00
Closing Cash Balance	90,000.00	20,000.00	-21,500.00

Cash from Operating Activities

Let's return to our company ABC and suppose they are a retail business selling clothing. The cash flow from operating activities can be seen as the net cash position of the company arising out of the core business.

It reflects revenue coming into the company, in the case of ABC this could be from the sales of clothing, and costs coming out of the company, for example rent on the store in which ABC sells their clothing.

Other operating activities that might be included in the cash flow statement:

- Interest payments
- Income tax payments
- Payments made to suppliers of goods and services used in production
- Salary and wage payments to employees
- Depreciation
- Change in investment portfolio

Cash from Investing Activities

Investing activities include any uses of the company's cash to purchase an asset or the cash coming in from the sale of an asset.

Say for example that ABC owns some property from which they lease out to other companies to use. They might decide to sell this asset for £1000 and purchase an £800 building in another area. Hence, the net cash flow from ABC's investing activities is $£1000 - £800 = £200$.

Other investing activities include:

- Any payments relating to mergers and acquisitions (M&A) of other companies

- Loans the company makes to suppliers

Cash from Financing Activities

This part of the cash flow statement includes all cash that passes from investors into the company, as well as cash out of the company to its shareholders.

A company could at the end of every quarter issue a dividend to its shareholders, this would represent cash out of the company through financing. They could also receive cash flows into the company through the sale of its stock.

Other financing activities include:

- Payments for stock repurchases by the company
- Repayment of debt to banks (from potential loans the company might require)
- Issuance of a corporate bond

Now we will look at how to calculate cash flow, there are two methods: the direct method and the indirect method.

Direct Cash Flow Method

This is a very straightforward way of calculating the cash flow of a company. You essentially just add up all the cash coming in and subtract all the cash going out to find the company's net position over a certain period.

By going through each aspect of the cash flow statement this is relatively easy and is more common amongst smaller businesses as it provides great insight as to the current position of the company and the best actions to move it forward.

Indirect Cash Flow Method

The other accounting method is indirect cash flow. This is different from the direct method in that you begin with the net income generated over a period (from the income statement), and this value is then added or subtracted based on the assets and liabilities located in the balance sheet.

The main difference between the direct and indirect method is when it comes to cash flows from operating activities. In the direct method this is treated as normal cash flows in and out of the business, but in the indirect method you start with net income and then change the cash flow based on the change in assets and liabilities. Both methods should yield similar results, although most accountants prefer the indirect method as it can be simpler to prepare.

Limitations of the Cash Flow Statement

The cash flow statement is very useful in understanding the movement of cash within a company, however it can be misleading when being used to compare companies within the same industry.

As cash flow statements don't normally show a breakdown of the costs and revenues of a company, it can be difficult to interpret the profitability of said company. Hence, comparing cash flows of two different companies to determine which one is more profitable can be a mistake.

Also, since the cash flow statement only shows the cash position of the company at the end of a period, it is not a true reflection of the company's liquidity, as it could own some liquid assets not displayed in the statement.

Income Statement

An income statement is a financial report that details a company's income and expenses usually over each quarter.

The income statement is defined by the following equation:

$$\text{Net Income} = (\text{Total Revenue} + \text{Gains}) - (\text{Total Expenses} + \text{Losses})$$

The purpose of the income statement is to show where a company makes its revenues and incurs its expenses, and is vital in analysing the health of a company.

The revenue can be further divided into operating and non-operating revenue, this depends on the specific business model of the company, but dividing the revenue like this can provide greater clarity on where the company is generating its revenue and where it can potentially expand.

Income statement

ABC Limited

Q1 2022

Financial statements in US Dollars

Operating Revenue

Gross sales	5,000,000.00
Revenue of goods sold	5,000,000.00

Operating Expenses

Cost of goods	500,000.00
Employee Salary	400,000.00
Rent	600,000.00
Sales returns and allowances	150,000.00
Cost of goods sold	1,650,000.00
Net operating income	3,350,000.00

Non-operating Revenue

Interest	150,000.00
Rent	250,000.00
Non-operating revenue	400,000.00

Non-operating Expenses

Advertising	100,000.00
Amortisation	50,000.00
Bank charges	50,000.00
Charitable contributions	10,000.00
Commissions	50,000.00
Depreciation	40,000.00
Insurance	75,000.00
Interest	30,000.00
Non-operating expenses	405,000.00
Net non-operating income	-5,000.00

Other income

Gains on sale of assets	30,000.00
Loss of changing location	1,000,000.00
Total other income	-970,000.00
Net income	2,375,000.00

Operating Revenue

This is revenue gained by the company through its primary activities. Let's take our example of our retail company ABC. Their operating revenue would largely consist of revenue gained from the sales of their clothing.

Non-operating Revenue

This is revenue gained through secondary activities from the company. If we look at ABC, this could include things like interest earned on business capital in a bank account, or rent from other properties that ABC owns.

Gains

This is subtly different from non-operating revenue, in that gains usually represent non-business activities that are quite irregular. For example, ABC might be upgrading their delivery system for online orders and could be upgrading their transport vans. The revenue received from selling their old vans would be seen as a gain.

Similar to revenue, expenses are split into operating expenses and non-operating expenses, where operating expenses are those incurred from the primary activities of the business. Non-operating expenses are those that come about from alternative business activities.

Operating Expenses

An example of this for ABC could be the cost of purchasing clothing goods from their supplier.

Non-operating expenses

In the case of ABC, this could be the cost of interest that needs to be paid on a loan from a bank.

Losses

This is sometimes included in non-operating expenses although can be slightly different. Just like gains these are irregular, usually one-off losses accrued by the company. An example of a loss for ABC could be the cost of changing the location of some of their business operations.

Limitations of the income statement

The income statement is an incredibly useful document, however it is by no means the sole document in determining a successful company. Something to beware about income statements is they tend to vary widely over different time periods. This is because income statements will include one-off transactions that could drastically change the net income for that period, but is not reflective of the long-term net income of the company.

Also, the income statement does not provide any detail on the actual business model. How the company generates revenue and tries to cut costs. It is for this reason that when trying to value companies, it is important to look at not just the quantitative data as derived from financial statements, but also the qualitative aspects.

PE ratio

The PE ratio is one of the fundamental metrics used to analyse companies as a possible investment opportunity.

It is the price-to-earnings ratio and it is a ratio between a company's share price and its earnings per share. Now, what do these values actually mean? A company's share price is the price of one share on the stock market. If we go back to our example of ABC, its share price is the current price of one share on the London Stock Market.

Earnings per share is a bit more difficult to obtain. To calculate this you need the earnings the company has generated (usually over the past 12 months) and the total amount of shares issued by the company. 'Earnings' is just another word for net income, so you can find the total earnings of a company by looking at their income statement as previously discussed.

To find the total number of shares issued, most financial websites should have this available on their website. Note, some companies might report their earnings per share themselves on their income statement, however it is still good to know how to calculate this yourself.

For example, say ABC reported earnings of £1 million over the past 12 months and they have 100,000 shares outstanding. Then their earnings per share is $\frac{£1,000,000}{100,000} = £10$. Now suppose the current price of one share is £20 on the stock market, then ABC's P/E ratio is $\frac{£20}{£10} = 2$.

This is known as a trailing PE ratio, where earnings are based on previous earnings of the company.

Another form of PE ratio is forward PE ratio, where instead of using past earnings, you use projected earnings that the company might generate.

Why use PE ratio?

PE ratio can be seen as determining the value of a stock, and is usually compared to the company's previous PE ratios, or the PE ratios of other similar companies. If the PE ratio is unusually low, an investor may see the stock as being undervalued and hence might decide to long the stock. On the other hand if the PE ratio is unusually high then the stock might be overvalued and an investor might want to short the stock.

Limitations of PE ratio

Like all financial analysis, using PE ratios is not foolproof and there are particular limitations with using it. One is that it can't be used for companies that generate negative earnings, which is quite common especially for younger companies. Also, share prices are incredibly volatile, which could incredibly skew that value of your PE ratio. When using a trailing PE ratio, there is no guarantee that past performance will be an indicator of the company's future results. When using a forward PE ratio, there is a possibility that a company will not meet its forecasted earnings.

PS ratio

Similar to PE ratio, the PS ratio is another metric used to analyse companies and their profitability. It is the price-to-sales ratio and compares the price of one share of a company to its sales per share, usually over the past 12 months.

Again, you can find a company's share price on the exchange that it is listed on, and 'sales' is just another word for revenue generated. Hence, you can find the company's sales per share by taking the revenue generated on the income statement and dividing it by the total number of shares issued.

For example, say ABC reported revenues of £2.5 million over the past 12 months and they have 100,000 shares outstanding. Then their sales per share is $\frac{£2,500,000}{100,000} = £25$. Now suppose the current price of one share is £20 on the stock market, then ABC's PS ratio is $\frac{£20}{£25} = 0.8$.

Similar to PE ratio, this is a trailing PS ratio, where we used past revenue. We could also use forward PS ratio, where we use forecasted sales over the next year.

Why use PS ratio?

PS ratio is another metric that can help allude to the financial health of a company. Similar to PE ratio, a low value of PS ratio compared to other companies in its sector could be a signal of an undervalued stock, and a high PS ratio could show a company is overvalued. However, one benefit of PS ratio over PE ratio is that it can be used for all companies, as it doesn't take into account costs and hence whether a company actually makes any profit.

Limitations of PS ratio

However, this can lead to one great inaccuracy of PS ratio, in that it doesn't take into account what costs a company is generating. So if a company is projected to see an increase in sales, but its costs are projected to increase at a greater rate, then it might not be a good investment opportunity, however the PS ratio will not take this into account.

EV

Enterprise Value (EV) is a measure of a company's total value, including all the assets that it owns. It is important to know that this is different from equity value, which is just a measure of a company's market value in equity. This is calculated by taking the current share price and multiplying it by the total number of shares outstanding:

$$\text{Equity value} = \text{price per share} \times \text{number of shares outstanding}$$

This is also known as market capitalisation.

Enterprise value on the other hand is calculated by the following:

$$\text{Enterprise value} = \text{market cap} + \text{debt} - \text{cash or cash equivalents}$$

What does EV tell you?

When looking at Enterprise value for the first time it may just seem like some arbitrary calculation that would not be particularly useful to anyone. However, it is important to look at this from the perspective of a potential investor trying to evaluate what the real value of a company is.

We have already established what the market cap (equity value) is. Debt is what the company owes to its creditors (people that have loaned money to the company). Cash or cash equivalents are just the amount of cash or very liquid assets the company owns.

Suppose John would like to purchase ABC. In order to develop a fair value of the company he may try to determine the enterprise value of the company. Part of what he would have to pay for the company would be its equity value, as we already discussed. However, another cost he would have to pay for is the debt the company owes. We subtract cash or cash equivalents because this would reduce what John has to pay for ABC as he can use the cash to pay off some of the debt, for example.

EV is very useful as it takes into account not only market cap but also the debt a company owes and its cash in reserves, both of which can be found on the balance sheet document we already discussed.

EV can also be used for useful metrics such as EV/EBITDA and EV/Sales to aid in gaining better accuracy on the value of a company.

EBITDA

EBITDA stands for earnings before interest, taxes, depreciation, and amortisation and is a widely used metric for determining the profitability of a company.

We know earnings means the net income of a company. Interest refers to the payments a company is required to make on the debt it owes, when calculating EBITDA, we do not include the cost of these interest payments. Taxes are pretty self-explanatory; we do not include the cost of any taxes the company has to pay. Depreciation refers to the loss in value of physical (tangible) assets, such as machinery or vehicles, usually due to their use overtime in the company's business activities. Amortisation is similar to depreciation, except it refers to the decrease in value of intangible assets, such as patents on company products or software the company has.

$$EBITDA = \text{earnings} + \text{interest} + \text{taxes} + \text{depreciation} + \text{amortisation}$$

For example, let's say ABC has a net income of £1 million over the past 12 months. This is its earnings and can be obtained from the income statement. We will then add to this the costs of interest, taxes, depreciation and amortisation, as all these costs are subtracted from revenue in the income statement to obtain the earnings of £1 million. Suppose ABC paid £100,000 in taxes, £100,000 in interest on debt and lost £50,000 due to devaluation of its clothing machinery in its factories. Lastly, suppose ABC paid £200,000 for a patent on some of its manufacturing processes for 10 years, hence over the last year, it has incurred an amortisation cost of $\frac{£200,000}{10} = £20,000$.

So the EBITDA of ABC is $£1,000,000 + £100,000 + £100,000 + £50,000 + £20,000 = £1,270,000$.

Why use EBITDA?

This may seem like an odd metric, as we are just ignoring some of the costs incurred by the company. However, it is useful as it measures the company's value, minus any costs associated with tax and financing decisions, and how the depreciation of assets is scheduled. It is seen as having a loose similarity to cash flow and can be a good estimator of the profitability of a company. It is also useful for business owners in determining how much costs they can afford with relation to their company's structure and still remaining profitable.

EBITDA Disadvantages

Although, some investors reject the use of EBITDA as it can greatly alter the perception of a company's profitability. This is particularly stark in asset-intensive industries such as car manufacturing, where capital costs have an enormous effect on the business but are omitted in EBITDA.

Time Value of Money

The time value of money is an essential concept in understanding the idea of cash as an asset and evaluating potentially profitable investments. It is also intertwined with the idea of opportunity cost.

The time value of money is the idea that money in the present can increase in value in the future when it is invested into some sort of asset. As a result of this, most investors prefer to receive money in the present than in the future, as it gives you greater time, and hence a better opportunity to invest that money. Let's look at a common example to illustrate this.

Say you have the choice between receiving £100 now, or £130 in three years' time. The average person would choose the £130 in three years, as they see this as more money than the £100. However, the savvy investor should choose the £100 now, as they have three years to invest that money. Say they invest that £100 into an asset that returns 10% of their capital a year (not as unreasonable as you may think), then after three years their £100 would be worth = £133.10. Which is £3.10 more than the £130. This is the time value of money.

We can see from the example above that calculating the time value of money must include the initial amount of money to be invested, the rate of return over a fixed time period and the total time period over which the money is being invested. We see the future value of money can be calculated as such

$$FV = PV \times (1 + i)^n$$

Where FV = future value, PV = present value, i = interest / return rate per year and n = number of years. So, using this formula in our example

$$FV = £100 \times (1 + 0.1)^3 = £133.10$$

This is also illustrating the powerful effect of compound interest, where the interest / return gained after each period is increased, as the total capital being invested is now greater than in the period before. For example, £100 is invested so

After year 1, the total value of money is now $£100 \times 1.1 = £110 = £100 + £10$.

After year 2, £110 is invested, so the total value of money is now $£110 \times 1.1 = £121 = £110 + £11$.

After year 3, £121 is invested, so the total value of money is now $£121 \times 1.1 = £133.10 = £121 + £12.10$.

So, we can see the interest payments increase each year.

Geometric Series

Before we move onto the following two chapters, it is worthwhile reviewing some maths that will be useful, in particular geometric series.

Suppose we had a series of the following format:

$$a + ar + ar^2 + ar^3 + \dots + ar^{(n-1)}$$

We say that this series has a starting term a , and a common ratio r . Also note it has n terms.

When n becomes very large, it can be quite cumbersome to calculate all these terms then sum them up. Luckily for us, there is a simple formula which we will prove that will allow us to calculate this expression much more quickly.

Let us define this sum as such

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{(n-1)}$$

If we multiply both sides of the equation by r that will give us

$$S_n r = ar + ar^2 + ar^3 + ar^4 + \dots + ar^n$$

If we subtract these two equations from each other we get the following

$$S_n r - S_n = (ar + ar^2 + ar^3 + ar^4 + \dots + ar^n) - (a + ar + ar^2 + ar^3 + \dots + ar^{(n-1)})$$

$$\begin{aligned} S_n(r - 1) &= ar^n - a \\ &= a(r^n - 1) \end{aligned}$$

$$S_n = \frac{a(r^n - 1)}{(r - 1)}$$

$$\text{So, } a + ar + ar^2 + ar^3 + \dots + ar^{(n-1)} = \frac{a(r^n - 1)}{(r - 1)}$$

So as long as we know the first term, common ratio and number of terms, we can now find the value of any geometric series.

Sum to infinity

Now suppose that the common ratio r of the geometric series is between -1 and 1 , i.e. $|r| < 1$.

Then that means that as we continue to raise the power of r it will tend to zero. In other words, $\lim_{n \rightarrow \infty} r^n = 0$ for $|r| < 1$.

Therefore, each subsequent term in the series

$$a + ar + ar^2 + ar^3 + \dots$$

Is getting smaller and smaller and hence there must be a limit of the sum, in other words

$\lim_{n \rightarrow \infty} a + ar + ar^2 + \dots + ar^n = S_{\infty}$ where S_{∞} is some real number. We can prove a formula that we use to calculate this limit.

From the previous proof we know

$$S_n = a + ar + ar^2 + ar^3 + \dots + ar^{(n-1)} = \frac{a(r^n - 1)}{(r - 1)}$$

Now suppose $|r| < 1$ and $n \rightarrow \infty$, then

$$S_{\infty} = \frac{a(-1)}{r-1}, \text{ as we know } \lim_{n \rightarrow \infty} r^n = 0.$$

$$S_{\infty} = \frac{a}{1-r}, \text{ is the limit of the geometric series}$$

This is known as the sum to infinity and is only true for $|r| < 1$.

Discounted Cash Flow

Further developing the idea of the time value of money, discounted cash flow analysis helps determine the current value of an asset based on its future cash flows.

Just like how we can use the time value of money to determine the value of money in the future given present value, we can also work backwards and determine the present value of money depending on the future value and the rate of return.

Suppose I wish to buy some property for my rental business, and based on some analysis this particular building will be worth £100,000 next year. I am also hoping to get a 5% return on my investment in this property, which can be based on forecasted growth rates in the property market. In order to determine what I should pay for this property I do the following calculation

$$\frac{£100,000}{1.05} = £95,238.10 \text{ which is what I should pay for now.}$$

This is the key concept to DCF analysis.

Now let's look at a more complex example. Say John is looking to purchase 20% of ABC shares. By looking at its cash flow statement you see that it generated £1 million in cash flow, so 20% of this would cost £200,000. Let's say that based on some analysis ABC is predicted to increase its cash flow by 10% year-on-year. So, after one year its cash flow will be $£200,000 \times 1.1 = £220,000$. After 2 years it will be $£220,000 \times 1.1 = £242,000$ and so on.

Now suppose you wish to obtain a 15% compound return on your investment for five years of owning 20% of ABC. That is for each future year of owning ABC you wish to earn a 15% return on that year. To determine how much you should pay now for that return, you do what we showed in the example, so for the first year you should pay and for the second year you should pay

So, in order to find out how much John would like to pay for 20% of ABC shares, obtaining a 15% compound return over 5 years we obtain:

$$DCF = \frac{£200,000 \times 1.1}{1.15} + \frac{(\£200,000 \times 1.1)^2}{1.15^2} + \frac{(\£200,000 \times 1.1)^3}{1.15^3} + \frac{(\£200,000 \times 1.1)^4}{1.15^4} + \frac{(\£200,000 \times 1.1)^5}{1.15^5} = £876,884.34$$

for a total future cash flow of

$$£200,000 \times 1.1 + £200,000 \times 1.1^2 + £200,000 \times 1.1^3 + £200,000 \times 1.1^4 + £200,000 \times 1.1^5 = £1,343,122$$

We can generalise this process for DCFs:

$$DCF = \frac{CF_1}{1+r} + \frac{CF_2}{(1+r)^2} + \dots + \frac{CF_n}{(1+r)^n}$$

Where DCF = discounted cash flow, CF_i = forecasted cash flow during period i ($i = 1, 2, 3 \dots n$), r = rate of return for each time period and n = total number of time periods.

Benefits of DCF

Discounted Cash Flow is a really useful analysis tool, as it ignores all market perceptions about the value of a stock and instead tries to predict the future value of a company based on objective data. It is also really good for estimating the long-term returns of an investment.

Disadvantages of DCF

DCF can be very inaccurate if used in isolation, just like all other analysis tools. This is mainly because DCF might not take into account the state of the economy or in particular the sector in which the company operates in.

Gordon Growth Model

The Gordon Growth Model (GGM) is similar to DCF analysis but instead uses the dividends per share issued by the company. The GGM assumes that a company's shares are worth the sum of all its future dividends discounted back to their current value. This is not too far fetched, because a company can only issue dividends if it has surplus profit, hence a company can only issue greater dividends if its profits increase, likely leading to an increase in share price. It also assumes a constant rate of growth for dividends per share that are issued.

You can find the dividends issued by a company on the cash flow statement and hence the dividends per share by dividing this value by the current number of shares outstanding. Suppose the dividends per share of ABC is predicted to grow at a constant rate of 5% per year, however you want a return of 10% a year when investing in this company.

So, to find the intrinsic value of one share suppose the share is currently trading at £180 with a dividend per share of £10. If I purchase the share now then I will expect in a year to receive a dividend payout of $£10 \times 1.05 = £10.50$. This is because I expect the dividends per share to increase by 5%, but I then have to discount this by my discount rate to find what the present value of this dividend payment is, similar to the discounted cash flow. So the present value of £10.50 in a year is $\frac{£10.50}{1.1} = £9.54$. This is what I should pay now for my dividend payment in a year's time.

Now after two years the dividend payment will increase to $£10 \times 1.05^2 = £11.025$, and to discount this to present day value according to the rate of return I want, I get $\frac{£11.025}{1.1^2} = £9.11$.

So in return for 2 years worth of dividend payments I should pay now

$$\frac{£10 \times 1.05}{1.1} + \frac{£10 \times 1.05^2}{1.1^2} = £9.54 + £9.11 = £18.65$$

Some of the more mathematically inclined reading this would have noticed this is a geometric series, but more importantly that this is a geometric series that tends to a limit, as each successive addition becomes smaller.

Now suppose we are going to own ABC shares for an infinite amount of time, this means we are going to receive an infinite amount of dividend payments. We can find what the present value of these dividend payments are by calculating the following series

$$\begin{aligned} PV &= \frac{£10 \times 1.05}{1.1} + \frac{£10 \times 1.05^2}{1.1^2} + \frac{£10 \times 1.05^3}{1.1^3} + \dots \\ &= \frac{£10 \times 1.05}{1 - \frac{1.05}{1.1}} \text{ (according to the sum to infinity formula)} \\ &= £190.91 \end{aligned}$$

Now, going back to our assumption that the sum of all future dividend payments discount to the current share value, then the intrinsic value of ABC is £190.91 per share, which is greater than the current market price of £180 per share. There the GGM suggests ABC is undervalued and could be a good investment.

So is the Gordon Growth Model actually useful?

The great thing about the GGM is that compared to a lot of other financial models it is not very complicated and is relatively easy to understand and apply to many different businesses. It is pretty useful when applied to more mature businesses who are likely to issue dividends and increase the amount of dividend they issue as well.

However, there can be some pretty serious issues when using the GGM and the most obvious of these is that it cannot be used on companies that don't issue any dividends, making it completely useless for growth stocks.

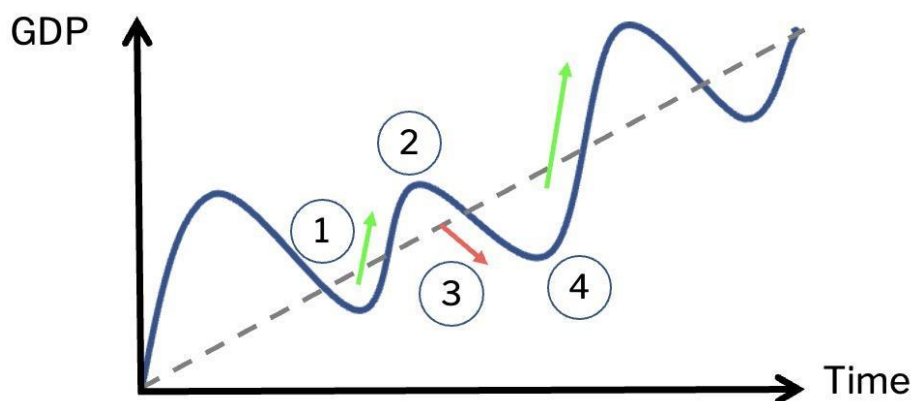
Also, the assumptions made when using it can be pretty detrimental, one it assumes constant dividend growth, and the accuracy of this dividend growth depends heavily on the analysis used to reach this figure.

Another issue is if the rate of return required by the investor is not greater than the expected growth rate of the dividends, then the common ratio is greater than 1, meaning our series of dividend payouts will tend to infinity,

giving the company an infinite value. In practice, this is meaningless and so we cannot use the GGM in this case.

Economic Cycle

The economic cycle is the theory that all economies around the world go through cyclical periods in terms of their growth rate. The economic cycle can be split up into four stages: expansion, slowdown, contraction and recovery. This is important in the world of equities, as the value of equities can broadly follow the growth rate of an economy.



There are different ways of measuring an economy's growth with the most popular being GDP (Gross Domestic Product). This is a measure of the total value of goods and services produced by an economy over a fixed period of time. Many factors affect GDP such as interest rates, inflation, unemployment rate and much more. However, depending on what stage in the economic cycle we are in we can sometimes determine what sectors are likely to perform well. During periods of high economic growth, sectors such as tech tend to perform strongly as these companies are usually leading economic growth due to their high innovation. During recessions or periods of economic decline, sectors such as water and electricity usually perform better as demand for more risky stocks decreases and investors move into more stable companies.

We could also look at seasonal cycles, where months such as September and January are historically the worst performing for stocks, but the run-up to Christmas seems to be a particularly profitable time to enter the stock market.

How to make money from equities

There are two main ways to make money from equities. This is not accounting for all the different strategies an investor or trader might employ depending on the company, its sector or the period we are in in the economic cycle. Instead, this looks at what the investor is looking for in terms of return from his/her investment.

Growth stocks

Growth stocks are stocks that are predicted to experience high levels of growth in its share price. This is the standard way you would invest in any asset. Buy the shares now in the expectation that in the future you can sell it for more than you bought it for.

There are particular stocks which tend to have a higher growth rate than other sectors. In more recent years these have been healthcare and tech stocks for example. These stocks usually come with a higher risk of the

company not performing as expected, but this is balanced with a greater potential return on your investment. Your investment could be based on some of the ratios and investment models we have previously discussed.

Dividend stocks

Dividend stocks are a form of fixed income investment where you receive income at regular time intervals. A dividend is a distribution of profits generated by a company to its shareholders, hence if you hold shares in a company and they decide to issue shares, then you are entitled to those profits. Where younger companies might be more interested in investing any surplus profits back into the company to stimulate growth, more mature companies tend to distribute those profits in the form of dividends to maintain demand for their stock. These are usually distributed at the end of every quarter.

Dividend yield is calculated by dividing the dividend received per share divided by the price of one share. It can be interpreted as the return in dividends for every £1 invested in the company. For example, say ABC distributes £0.50 per share every quarter, so over a year if you owned one share you would receive £2 in dividends (dividend yield is usually calculated over a year). Suppose the current share price is £20, then the dividend yield for the past year would be $\frac{£2}{£20} = 0.1$ or 10%. So, for every £1 invested in ABC you would receive $£1 \times 0.1 = £0.10$ in dividends per year.

Dividend stocks are usually seen as less risky as growth stocks due to the more regular periods of income, and hence are likely to produce less return. One important thing to note about dividend stocks is a dividend yield that is too high could be a bad thing, as this could lead to depreciation in the share price. Therefore, the amount of dividend received could be cancelled out by the drop in value of the stock.

Funds

A fund is a collection of different stocks in a portfolio. Different types of funds can suit different types of investors in terms of their risk tolerance and what they expect in terms of return, as well as their personal interests. Funds don't have to only contain stocks, but can also contain other assets such as cryptocurrencies, bonds and commodities.

The usual benefit of a fund is that it diversifies your portfolio across different sectors, geographical areas or perhaps even assets, spreading your risk and diversifying your portfolio compared to just investing in one stock.

Actively-Managed Funds

Actively-managed funds are managed by a fund manager. The fund manager is responsible for choosing the allocation of different securities within the overall fund. A fund might have a certain profile or objective that the fund manager is trying to obtain. The fund might be specialising in aviation stock across the world, or emerging markets in Europe. It could even just focus on real estate in Brazil. It is up to the fund manager and up to the investor whether this fund is in alignment with what you want to invest in.

Actively-managed funds typically have higher costs associated with them as you are paying for the fund manager to alter the funds allocations based on market conditions and other factors. Not only are you betting on the fund, but also the manager to make the correct decisions to essentially 'beat the market'.

Passive/Index Funds

Passive funds are the opposite to actively-managed funds. Rather than having a fund manager alter the allocations of the fund, passive funds usually track some sort of stock market index.

One of the most popular indexes is the S&P 500 which tracks the 500 largest funds in the USA based on market capitalisation. Another is the FTSE 100 which are the 100 largest companies in the UK again based on market capitalisation. The purpose of index funds is to essentially track a certain market, trying to match its returns. For this reason, index funds are seen as a lower risk option compared to actively-managed funds.

Passive/Index funds also tend to have lower costs as it is essentially just an algorithm tracking an index, hence there is no need to have a fund manager. Index funds can also be quite diverse, ranging from different geographic areas to different sectors.

Exchange-traded funds

An exchange-traded-fund (ETF) can be an active or passive fund. What defines an ETF is how it is traded. ETFs are like any other stock and are listed on the stock market. Hence, they can be traded at any time during trading hours and hence their price fluctuates throughout the day.

You can have ETFs that are either index funds or actively-managed funds, however it is more common to have them as index funds as the value of the index changes throughout the day and the lower costs associated with ETFs.

Mutual funds

These are the opposite of ETFs. Typically, index mutual funds can only be traded at certain points in a day, so whilst you might initiate a trade at some point during the day, it will only be executed at a specific time, usually based on the day's closing price. Mutual funds are controlled by a broker or an investment company, and it is here that you trade mutual funds.

You can also have mutual funds as either index or actively-managed funds. However, it is more likely to have them as actively-managed funds as the investment company pools funds from many individual investors to create the fund, giving the fund manager greater control.